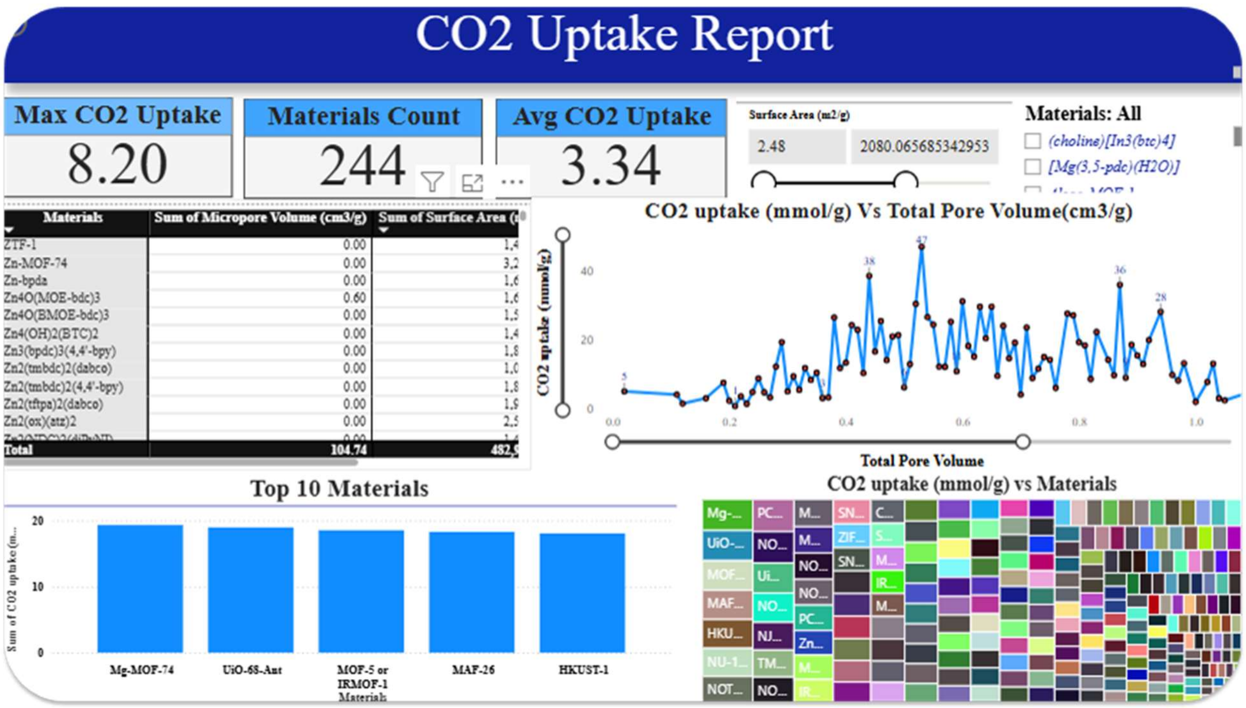
Designing Better Materials for CO_2 Capture



Introduction

Carbon dioxide capture is a critical challenge for sustainable energy and climate mitigation. Traditional experimental approaches to developing new materials are costly and time-consuming. In this project, we combined data analytics, machine learning, and interactive visualization in Power BI to uncover the key factors driving CO_2 uptake and to guide the design of more effective sorbent materials.

Step 1: Data Preparation

- Imported data from `datapoints.xlsx` into Power BI.
- Cleaned the dataset by removing duplicates, renaming columns for clarity, and replacing missing values (e.g., Micropore Volume filled with column average).
- Created DAX measures to calculate averages, ratios (e.g., Uptake per Area, Uptake per Pressure), and top-performing materials.

Step 2: Understanding Key Drivers of CO_2 Uptake

We first explored which features most strongly affect CO_2 capture performance:

- A correlation heatmap revealed that Surface Area and Pore Volume are strongly linked with higher uptake.
- Line plots showed that uptake increases with Surface Area up to a threshold, while pressure becomes the dominant factor at higher operating conditions.
- Slicers allowed filtering by specific materials to examine how each responds to pressure and temperature.

Insight: CO_2 uptake is not determined by a single property; it is the interaction of structural properties (surface/pore volume) and operating conditions (pressure/temperature) that defines performance.

Step 3: Identifying the Best Materials

Next, we asked: *Which materials perform best under different scenarios?*

- A ranked table and bar chart identified the Top 5 materials with the highest CO_2 uptake.
- Tree maps highlighted the contribution of each material to total uptake.
- By applying slicers:
  - At low pressures (post-combustion), materials A and B showed superior performance.
  - At high pressures (pre-combustion), materials C and D were most effective.
  - Some materials also demonstrated greater thermal stability, making them promising for industrial use.

Insight: Different materials are optimal depending on the capture scenario (low vs. high pressure).

Step 4: Linking Chemistry to Performance

We also examined chemical composition:

- Grouping materials by metal centers showed that transition metals generally enhance uptake.
- Functional group analysis suggested that nitrogen-containing groups improved CO_2 affinity.

Insight: Structural factors alone cannot explain uptake; chemical composition plays a decisive role in performance.

Step 5: Machine Learning for Prediction

To move beyond descriptive analytics, we trained three ML models (ANN, LSTM, CNN-LSTM) to predict CO_2 uptake from surface area, pore volume, pressure, and temperature.

Model	*R^2 Train*	*R^2 Test*	*RMSE Train*	*RMSE Test*
ANN (Keras)	0.932	0.838	0.383	0.651
LSTM	0.963	0.789	0.282	0.742
CNN	0.969	0.814	0.257	0.697

- ANN provided the best balance between bias and variance, delivering strong generalization on unseen data.
- The Actual vs Predicted plots in Power BI confirmed the reliability of predictions.

Insight: ML models can accurately predict uptake and guide material design before experimental synthesis, saving both time and cost

Final Story: From Data to Material Design

Through this integrated analysis, we:

1. Identified key drivers (surface area, pore volume, pressure).
2. Ranked and compared materials under different operating conditions.
3. Linked chemistry to performance for rational design insights.
4. Applied ML to predict uptake and accelerate discovery of new sorbents.